

Systemic Tail Episode Risk: Persistence and Cumulative Severity of Synchronized Extreme-Loss Episodes Across Markets

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This paper extends Tail Episode Risk (TER) and Conditional Tail Episode Risk (CTER) — introduced in the author’s companion paper on path-dependent tail risk — to the multivariate, cross-market setting. Preliminary working paper circulated for comment. Replication code available from the author.

Abstract

Systemic-risk measures such as CoVaR, marginal expected shortfall, and co-exceedances quantify the *magnitude of contemporaneous joint tail stress*; a related literature measures the *timing* of spillover. None targets the *persistence and cumulative severity of a synchronized stress episode*. We introduce **Systemic Tail Episode Risk (STER)**, the conditional upper-tail quantile of the worst contiguous, breadth-gated, cross-asset threshold-exceedance episode over a forecast horizon, and its conditional extension **CSTER**. STER is a multivariate max cluster functional; we establish that it reduces to univariate TER in one dimension, is monotone in the breadth gate, and separates loss paths that existing point-in-time systemic measures rank identically. In simulation, STER rises sharply with cross-asset tail dependence while portfolio VaR and ES stay essentially flat — the cross-sectional analogue of TER’s response to serial dependence. Empirically, on five markets over 2002–2026, realized STER isolates exactly the broad, sustained crises one would want a systemic measure to flag (COVID-2020, the 2008–2009 global financial crisis, the April 2025 selloff, August 2015, and Brexit), in economically sensible order. As with TER, STER is elicitable while the CSTER product is not, implying component-wise scoring. Consistent with the companion paper, we separate the innovation of the risk functional from its conditional-forecasting value, which we specify and leave to empirical companion work.

Keywords: Systemic risk; tail risk; cross-asset contagion; path-dependent risk; extreme-value theory; CoVaR.

JEL Classification: C58; C53; G01; G17; G32.

1 Introduction

The companion paper introduces Tail Episode Risk (TER), $\text{TER}_{\alpha,\beta,H,t} = Q_{\beta}(S_{t,H}^* \mid \mathcal{F}_t)$, the conditional upper-tail quantile of the worst contiguous cumulative threshold-exceedance episode $S_{t,H}^*$ over a horizon, and shows that it captures a temporal-organization dimension of tail risk that marginal Value-at-Risk (VaR) and Expected Shortfall (ES) discard. That framework is univariate. Its own adversarial stress tests, however, flag a limitation it cannot resolve: cross-asset contagion is

inherently multivariate, and a univariate episode functional cannot express the risk that *several* markets enter sustained stress *together*.

This paper supplies that extension. The economically important systemic event is not a single deep cross-sectional shock, nor a single market’s prolonged decline, but a *synchronized and persistent* deterioration — many markets breaching their tails jointly, and staying there. Existing systemic-risk measures do not target this object. CoVaR (Adrian and Brunnermeier, 2016), marginal expected shortfall and SRISK (Acharya et al., 2017; Brownlees and Engle, 2017), co-exceedance counts (Bae et al., 2003), and their multivariate and network variants all quantify the *magnitude of contemporaneous joint tail stress*; a distinct strand measures the *timing* of spillover, such as the duration-weighted spillover-persistence statistic of (Spillover persistence working paper). The persistence-and-cumulative-severity of a joint episode falls between them.

We introduce **Systemic Tail Episode Risk (STER)**. Fix per-asset tail thresholds and, at each step, count the *breadth* of joint exceedance; a *systemic episode* is a contiguous run in which breadth meets a gate κ , and its severity is the cumulative cross-asset excess over the run. STER is the conditional upper-tail quantile of the worst such episode over the horizon; its conditional extension CSTER decomposes systemic episode risk into an occurrence probability and a conditional severity. The construction deliberately reuses the univariate machinery (thresholds $\rightarrow$ episodes $\rightarrow$ worst-episode severity $\rightarrow$ tail quantile), adding one dimension: cross-market breadth.

We make four points, all established below without overclaiming. First, STER is honestly located: as an object it is a *multivariate max cluster functional* (Basrak and Segers, 2009), and we claim the financial, conditional, breadth-gated, forecast-oriented *application*, not the object. Second, STER is distinct from point-in-time systemic measures: we construct loss paths with identical per-asset and portfolio marginals and identical co-exceedance counts — so identical portfolio VaR/ES, CoVaR, and MES — but different STER. Third, simulation shows STER rises with cross-asset tail dependence while portfolio VaR/ES stay flat. Fourth, realized STER isolates the broad, sustained historical crises. As in the companion paper, we separate *measure innovation* from *forecasting performance*: the conditional-forecasting evaluation of CSTER is specified here and carried out in companion empirical work, where — given the rarity of systemic episodes — we expect it to be, at best, modest.

2 Related literature and positioning

Point-in-time systemic measures. CoVaR and Δ CoVaR (Adrian and Brunnermeier, 2016) measure the system’s VaR conditional on an institution’s distress; marginal expected shortfall and SRISK (Acharya et al., 2017; Brownlees and Engle, 2017) measure an institution’s expected loss when the system is in its tail; co-exceedances (Bae et al., 2003) count simultaneous tail breaches; multivariate and network extensions (e.g. multivariate CoVaR, tail-risk networks) enrich the cross-section. All evaluate joint tail stress *at a point in time*: they are functions of the joint loss distribution on a single date and are invariant to the temporal ordering of dates. STER is, by construction, not (Proposition 3).

Temporal systemic measures. A smaller literature adds a time dimension. Spillover persistence (, Spillover persistence working paper) measures the duration-weighted average *lag* between an institution’s loss and the system’s — a Macaulay-duration analogue of transmission timing — not the cumulative severity of a contiguous joint-stress run. Duration-based extreme-value models study waiting times *between* extreme events. STER differs from both: it measures how severe the worst *sustained, synchronized* episode becomes, not how long transmission takes or how long one waits between shocks.

Multivariate extremes. The probabilistic object underlying STER — a functional of a cluster of joint exceedances of a multivariate regularly-varying series — belongs to the theory of multivariate tail processes and cluster functionals (Basrak and Segers, 2009). Exactly as the univariate TER is a cluster functional in the sense of Segers (2003), STER is a multivariate max cluster functional. We therefore claim novelty only in the financial, conditional, breadth-gated, forecast-oriented formulation, not in the underlying functional.

3 From TER to Systemic Tail Episode Risk

3.1 Breadth-gated systemic episodes

Let L_{t+k}^i be the loss on asset $i \in \{1, \dots, d\}$ at step k of the horizon H , and let $q_{t,\alpha}^i$ be its conditional tail threshold (the trailing $(1 - \alpha)$ -quantile of L^i). Define the per-step *breadth* and the weighted cross-asset excess

$$b_{t+k} = \sum_{i=1}^d \mathbf{1}\{L_{t+k}^i > q_{t,\alpha}^i\}, \quad c_{t+k} = \sum_{i=1}^d w_i (L_{t+k}^i - q_{t,\alpha}^i)_+,$$

with portfolio weights $w_i \geq 0$, $\sum_i w_i = 1$. Fix a breadth gate $\kappa \in \{1, \dots, d\}$. A *systemic episode* is a maximal contiguous run of steps with $b_{t+k} \geq \kappa$; on such steps the episode accrues severity c_{t+k} , and on non-systemic steps it does not. For systemic episode j with index set $\mathcal{K}_j$, its severity is $S_{t,H,j}^{\text{sys}} = \sum_{k \in \mathcal{K}_j} c_{t+k}$, and the worst-episode severity over the horizon is

$$S_{t,H}^{\text{sys}*} = \max_j S_{t,H,j}^{\text{sys}} \quad (= 0 \text{ if no systemic episode occurs}).$$

3.2 Systemic Tail Episode Risk and its conditional extension

$\text{STER}_{\alpha,\beta,H,t} = Q_\beta(S_{t,H}^{\text{sys}*} \mid \mathcal{F}_t)$

is the conditional upper-tail quantile of the worst systemic episode. Defining the systemic-occurrence event $O_{t,H}^{\text{sys}} = \mathbf{1}\{S_{t,H}^{\text{sys}*} > 0\}$, the conditional extension is

$$\text{CSTER}_{\alpha,\beta,H,t} = \underbrace{\mathbb{P}(O_{t,H}^{\text{sys}} = 1 \mid X_t)}_{\text{systemic occurrence}} \cdot \underbrace{\text{ES}_\beta[S_{t,H}^{\text{sys}*} \mid O_{t,H}^{\text{sys}} = 1, X_t, Z_t]}_{\text{conditional systemic severity}},$$

the direct systemic analogue of CTER, separating the likelihood that markets *jointly* enter a sustained stress episode from its potential severity.

4 Theoretical properties

Throughout, $\mathcal{F}_t$ is given, so all thresholds are known and the statements are pathwise.

Proposition 1 (Reduction to TER). *For $d = 1$ and any κ (necessarily $\kappa = 1$), $S_{t,H}^{\text{sys}*} = S_{t,H}^*$ and hence $\text{STER}_{\alpha,\beta,H,t} = \text{TER}_{\alpha,\beta,H,t}$.*

Proof. With $d = 1$, $b_{t+k} = \mathbf{1}\{L_{t+k} > q_{t,\alpha}\}$ and $\kappa = 1$, so a systemic step is exactly an exceedance and $c_{t+k} = (L_{t+k} - q_{t,\alpha})_+$. Systemic episodes coincide with univariate episodes and their severities with $S_{t,H,j}$; the maxima and their conditional quantiles coincide. $\square$

Proposition 2 (Breadth monotonicity). *For every realization, $S_{t,H}^{\text{sys}*}$ is non-increasing in the breadth gate κ ; hence STER is non-increasing in κ .*

Proof. Raising κ can only turn systemic steps ($b \geq \kappa$) into non-systemic ones, never the reverse. Each episode’s index set under $\kappa' > \kappa$ is a subset of its set under κ with non-negative accruals c_{t+k} , so every episode severity is non-increasing, and so is their maximum. Monotone transformation of the conditional distribution preserves the quantile ordering. $\square$

Proposition 3 (Separation from point-in-time systemic measures). *There exist loss paths A, B that are temporal reorderings of one another — hence with identical per-asset marginals, identical portfolio-loss marginal (so identical portfolio VaR and ES, CoVaR, and MES) and identical co-exceedance counts — yet with $\text{STER}(A) \neq \text{STER}(B)$.*

Proof by construction. Take $d = 2$, $w = (\frac{1}{2}, \frac{1}{2})$, thresholds $q = (2, 2)$, $\kappa = 2$, and the six daily loss vectors $\{(4, 4), (4, 4), (4, 4), (0, 0), (0, 0), (0, 0)\}$. Path A orders them clustered, $A = ((4, 4), (4, 4), (4, 4), (0, 0), (0, 0), (0, 0))$; path B staggers them, $B = ((4, 4), (0, 0), (4, 4), (0, 0), (4, 4), (0, 0))$. Both use the same multiset of daily vectors, so per-asset marginals, the portfolio-loss marginal, and the number of co-exceedance days (three, each with breadth 2) are identical; any measure that is a function of the joint one-period distribution — portfolio VaR/ES, CoVaR, MES, co-exceedance counts — assigns them equal risk. But A forms one three-day systemic episode with severity $S^{\text{sys}*}(A) = 3 \times (\frac{1}{2} \cdot 2 + \frac{1}{2} \cdot 2) = 6$, whereas B forms three isolated one-day systemic episodes with $S^{\text{sys}*}(B) = 2$. Hence $\text{STER}(A) \neq \text{STER}(B)$. $\square$

Proposition 3 is the systemic counterpart of the univariate A/B example and is the core distinctness claim: STER responds to the *temporal organization* of joint stress that point-in-time systemic measures, by construction, cannot see.

5 Cross-asset tail dependence: simulation

We test whether STER responds to *cross-asset tail dependence* when the marginals and linear correlation are held fixed — the cross-sectional analogue of the companion paper’s serial-dependence experiment. We simulate $d = 6$ assets with standard-normal margins and linear correlation $\rho = 0.35$, and vary only the *tail* dependence through the degrees of freedom ν of a t -copula (small ν : strong joint extremes; large $\nu \rightarrow$ Gaussian: weak), reporting the upper tail-dependence coefficient λ . Portfolio VaR/ES are computed on the equal-weight portfolio; STER uses $\kappa = 2$, $H = 10$, threshold $\alpha = 95\%$, over 60,000 windows.

Table 1: As cross-asset tail dependence λ rises (with margins and linear correlation fixed), STER rises while portfolio VaR/ES stay essentially flat.

λ	Portfolio VaR 95%	Portfolio ES 95%	STER 95%	STER 99%
0.000	1.108	1.399	0.424	0.685
0.026	1.101	1.406	0.451	0.764
0.109	1.103	1.437	0.513	0.846
0.181	1.106	1.448	0.547	0.958
0.238	1.081	1.448	0.592	0.991

As λ rises from 0 to 0.24, STER99 increases by roughly 45% ($0.685 \rightarrow 0.991$) while portfolio VaR95 is unchanged (≈ 1.10) and portfolio ES95 rises only $\approx 3.5\%$. Cross-asset tail dependence materially raises systemic episode risk without proportionally raising portfolio marginal risk.

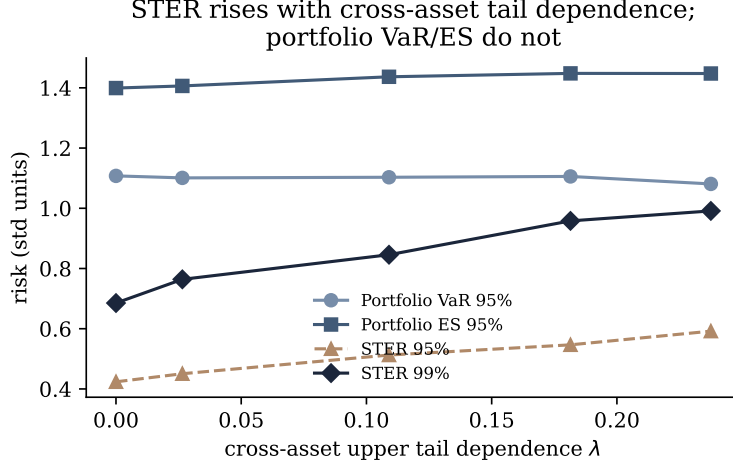


Figure 1: Simulation: STER (especially at 99%) rises with cross-asset upper tail dependence λ while portfolio VaR/ES remain flat.

6 Elicitability and scoring

The evaluation results of the companion paper transfer directly. $\text{STER} = Q_\beta(S_{t,H}^{\text{sys}*} \mid \mathcal{F}_t)$ is a conditional quantile of the observable, $\mathcal{F}_{t+H}$ -measurable random variable $S_{t,H}^{\text{sys}*}$; it is therefore *elicitable*, strictly consistently scored by the pinball loss, and admits a model-free coverage backtest $\mathbb{E}[\mathbf{1}\{S_{t,H}^{\text{sys}*} \leq \text{STER}\} \mid \mathcal{F}_t] = \beta$. The CSTER product $p_t m_t$ inherits the non-elicitability of the CTER product: the same convex-level-sets argument applies, so CSTER must be evaluated component-wise — systemic occurrence by a probability score, conditional systemic severity by a Fissler–Ziegel score on the systemic-episode subsample.

7 Empirical evidence: does STER flag systemic crises?

As an economic reality check we compute realized $S_{t,H}^{\text{sys}*}$ on five markets that span the 2008 crisis — the S&P 500, Nasdaq, the 20+ year Treasury ETF (TLT), EUR/USD, and USD/JPY — daily over 2002–2026 (common sample 6,012 observations), with $W = 1,250$, $H = 10$, $\alpha = 95\%$, $\kappa = 2$, equal weights. Table 2 lists the worst distinct systemic episodes.

Table 2: Worst distinct realized systemic episodes (cumulative breadth-gated joint severity, %).

Episode (date)	$S^{\text{sys}*}$ (%)
Mar 2020 (COVID)	8.34
Sep 2008 (Lehman / GFC)	6.18
Nov 2008 (GFC)	3.77
Apr 2025 (selloff)	3.59
Jan 2009 (GFC aftermath)	2.81
Aug 2015 (China / global selloff)	2.58
May 2022 (bear market)	2.01
Jun 2016 (Brexit)	1.81

STER isolates precisely the broad, sustained, cross-market crises, in economically sensible order: COVID-2020 dominates, followed by the Lehman peak and the wider 2008–2009 global financial crisis, the April 2025 selloff, the August 2015 global selloff, the 2022 bear market, and Brexit. Notably, episodes that are severe but *market-specific* rather than broadly synchronized do not top the systemic ranking — the intended behavior of a breadth-gated measure, and a point of contrast with the univariate TER ranking of the companion paper.

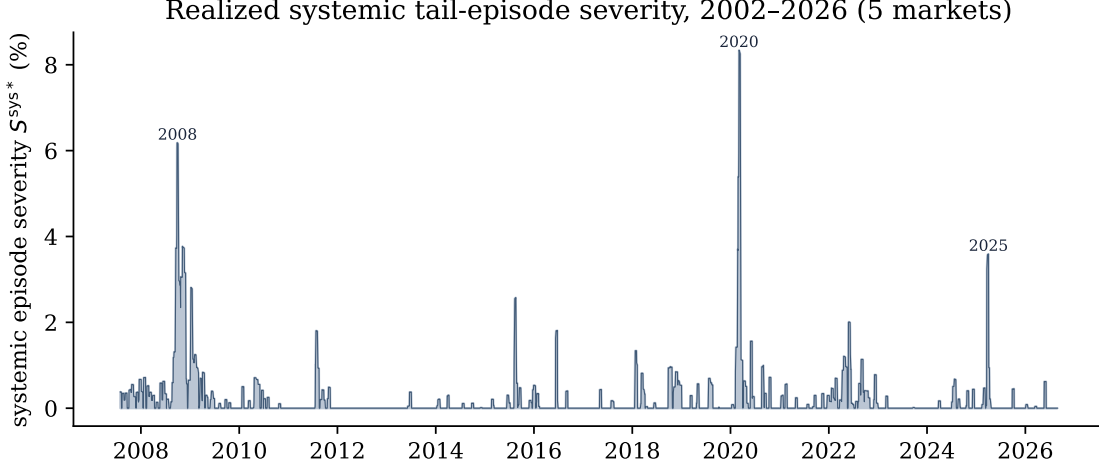


Figure 2: Realized systemic tail-episode severity across five markets, 2002–2026. The largest spikes coincide with the 2008–2009 crisis, the 2020 COVID crash, and the 2025 selloff.

8 Conditional forecasting: specification

CSTER poses the systemic forecasting question: how likely are markets to *jointly* enter a sustained stress episode, and how severe could it be? Its evaluation follows the companion paper exactly — systemic-occurrence discrimination by Brier score and AUC, conditional systemic severity by a Fissler–Ziegel score on the systemic-episode subsample, and an incremental test comparing a conditioned model against a parsimonious volatility/dependence baseline, all under a purged walk-forward protocol. We report this specification here and defer the empirical forecasting results to companion work, for a reason of scientific candor rather than convenience: systemic episodes are rarer than univariate ones, so the conditional layer will be estimated on thin data and, consistent with the companion paper’s univariate null, we do not anticipate a large incremental forecasting gain. As there, the contribution is the functional and its properties; the forecasting layer is reported honestly, whatever it shows.

9 Limitations

(i) *Prior art*. The underlying object is a multivariate cluster functional (Basrak and Segers, 2009); the contribution is the financial, conditional, breadth-gated formulation, not the functional. (ii) *Crowded neighborhood*. Systemic-risk measurement is densely populated; STER’s value rests on the temporal-persistence dimension and its sharp distinction from CoVaR/MES and from spillover persistence. (iii) *Design parameters*. STER adds a breadth gate κ to the threshold α and horizon

H ; robustness across (κ, α, H) is required and follows the companion paper’s grid methodology. (iv) *Sparse systemic events*. Deep-tail and conditional-forecasting estimates are data-limited. (v) *Static weights and thresholds*. Time-varying weights, liquidity-adjusted losses, and covariate-driven breadth gates are natural extensions.

10 Conclusion

Systemic risk has a temporal dimension that point-in-time comovement measures discard: extreme losses across markets can arrive synchronized and *persist*. Systemic Tail Episode Risk formalizes the severity of the worst such synchronized, sustained episode as a conditional quantile of a multivariate max cluster functional. We showed that STER reduces to TER in one dimension, is monotone in breadth, and — by construction — separates joint-stress paths that portfolio VaR/ES, CoVaR, MES, and co-exceedance counts rank identically; that it rises with cross-asset tail dependence while portfolio VaR/ES stay flat; and that realized STER isolates the broad, sustained historical crises. Whether the conditional extension forecasts these episodes better than a parsimonious baseline is, as in the univariate case, a separate and open question, to be answered honestly in companion empirical work.

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